GRAPH-THEORY ANALYSIS & MATHEMATICAL MODELLING FOR ROUTE OPTIMIZATION

PRESENTED BY: NITISH & MANISH

OVERVIEW:

THIS PROJECT INTRODUCES STUDENTS TO THE FUNDAMENTALS OF GRAPH THEORY AND ITS PRACTICAL APPLICATION IN MODELING AND OPTIMIZING PHYSICAL TRANSPORTATION CORRIDORS. THE FIRST PHASE FOCUSES ON BUILDING A STRONG THEORETICAL FOUNDATION BY COVERING KEY GRAPH CONCEPTS–SUCH AS WEIGHTED GRAPHS AND SPATIAL NETWORKS–ALONGSIDE CLASSICAL THEOREMS AND PATHFINDING ALGORITHMS LIKE DIJKSTRA'S, BELLMAN FORD'S ETC.

IN THE SECOND PHASE, THIS KNOWLEDGE IS APPLIED TO MODEL THE CONNECTIVITY BETWEEN MAJOR URBAN HUBS, SUCH AS THE ABC-XYZ CORRIDOR, SIMULATING VARIOUS ROUTES AND TRANSIT PATHS WHILE ACCOUNTING FOR REAL-WORLD VARIABLES LIKE DISTANCE, VARIOUS STATIONS. BY ANALYZING "INTERMEDIATE CONNECTIVITY," THE PROJECT AIMS TO BRIDGE MATHEMATICAL THEORY WITH PRACTICAL INSIGHTS INTO LOGISTICS AND ROUTE OPTIMIZATION, ASSESSING THE EFFICIENCY AND ROBUSTNESS OF REGIONAL TRANSPORT NETWORKS.

WEEK-WISE PLAN

WEEK 1: INTRODUCTION TO BASIC DATA STRUCTURES :

BASICS REVISION

STL(STANDARD TEMPLATE LIBRARY)

ARRAYS, VECTORS

STACKS & QUEUES

WEEK 2: BASICS OF GRAPH THEORY

WHAT IS A GRAPH?

TYPES OF GRAPHS

TERMINOLOGIES

REPRESENTATION OF GRAPH THEORY

WALKS, PATHS, CYCLES

ADJACENCY MATRIX & ADJACENCY LIST

WEEK 3: GRAPH TRAVERSAL AND CONNECTIVITY

INTRODUCTION TO GRAPH TRAVERSAL

DFS

BFS

APPLICATIONS OF BFS AND DFS

DIJKSTRA'S ALGORITHM

WEEK 4: ALGORITHMS RELATED TO GRAPHS

BELLMAN FORD SHORTEST PATH ALGORITHM

KOSARAJU'S ALGORITHM

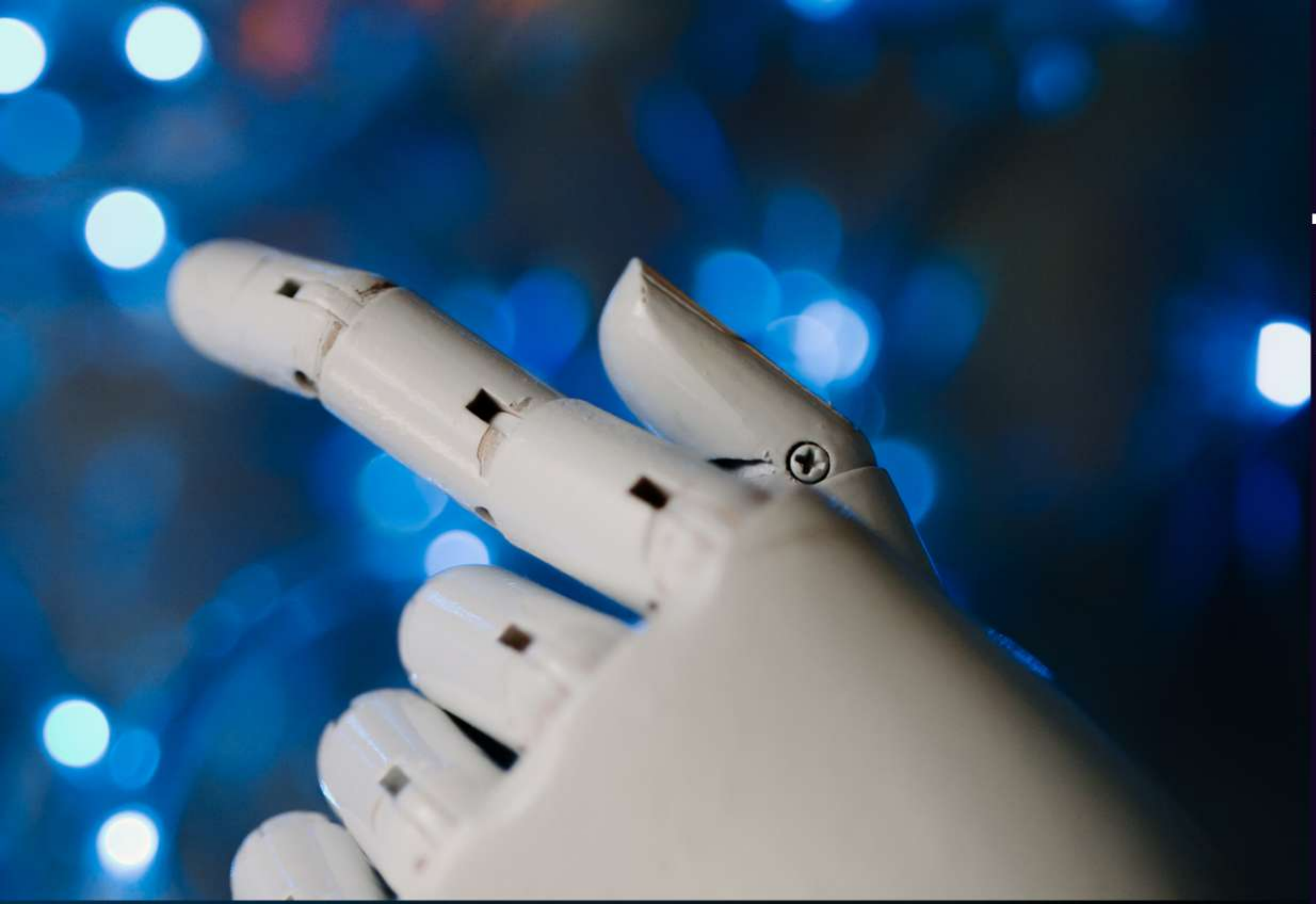
TARJAN'S ALGORITHM

WEEK 5: INTRODUCTION TO STOCHASTIC PROCESSES AND PROBLEM SOLVING

KRUSKAL'S ALGORITHM
INTRODUCTION TO STOCHASTIC PROCESSES
MISCELLANEOUS TOPICS
APPLICATIONS RELATED TO THESE ALGORITHMS
DISCUSSION OF USE CASES AND SOME PROBLEMS

WEEK 6-7: FINAL PROJECT AND EVALUATION

ROUTES OPTIMIZATION AND ANALYSIS OF PATHS
BASED UPON THE DISTANCE, AVAILABLE STATIONS, TIME ETC... USING THE GRAPH THEORY & STOCHASTIC PROCESSES



EVALUATION CRITERIA

REGULAR PARTICIPATION IN SESSIONS(~20%)
AND ASSIGNMENTS(~30%).
FINAL PROJECT (~50%).

PREREQUISITE

BASIC UNDERSTANDING OF FUNCTIONS,
AND LOGIC
BASIC KNOWLEDGE OF PROGRAMMING
(PREFERABLY IN C, C++, PYTHON OR R)


Thank You!